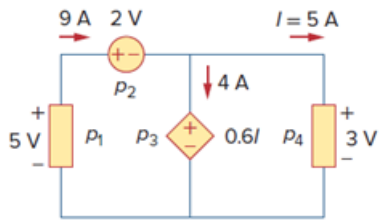


Problem

Compute the power absorbed or supplied by each component of the circuit in Fig. 1.16.

Figure 1.16:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

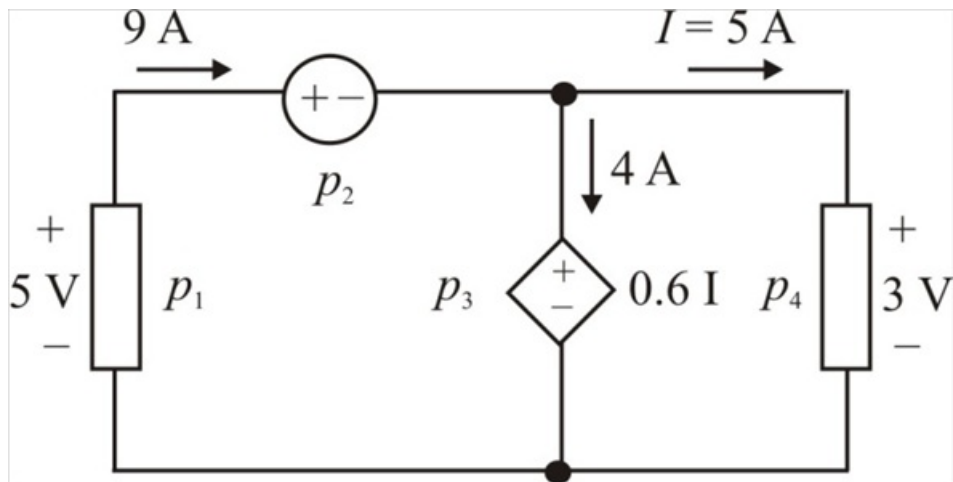


Figure 1

Step 2 of 5

The 9 A current is out of the positive terminal (or into the negative terminal),

Calculate the power p_1 .

$$\begin{aligned} p_1 &= vi \\ &= (5)(-9) \\ &= -45\text{ W} \end{aligned}$$

Therefore, the absorbed power is $\boxed{-45\text{ W}}$.

Step 3 of 5

The 5 A current is into the positive terminal (or out of the negative terminal),

Calculate the power p_4 .

$$\begin{aligned} p_4 &= vi \\ &= (3)(5) \\ &= 15 \text{ W} \end{aligned}$$

Therefore, the absorbed power is 15 W.

Step 4 of 5

The 9 A current is entering into the positive terminal (or out of the negative terminal) of the voltage source.

Calculate the power p_2 .

$$\begin{aligned} p_2 &= vi \\ &= (2)(9) \\ &= 18 \text{ W} \end{aligned}$$

Therefore, the power supplied by the independent source is 18 W.

Step 5 of 5

For p_3 , the 4 A current is into the positive terminal (or out of the negative terminal) calculate the power p_3 .

$$\begin{aligned} p_3 &= vi \\ &= (0.6I)(4) \\ &= [(0.6)(5)](4) \\ &= 12 \text{ W} \end{aligned}$$

Therefore, the power supplied by the dependent source is 12 W.